Saee Desai

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Summary

M.S. Data Science graduate with hands-on experience building and evaluating **LLM applications, agentic RAG systems, ML serving infrastructure, and multimodal generative AI**. Experienced with **Python, FastAPI, LangGraph, PyTorch, Docker, AWS, retrieval systems, model evaluation, observability, and reliability testing**. Two first-author publications in **post-quantum security and applied cryptography**.

Education

University of Maryland, College Park — Maryland, USA
***M.S. in Data Science**, GPA: 3.8/4.0* — *Aug 2024 – May 2026*

Vishwakarma Institute of Information Technology — Pune, India
***B.Tech in Computer Engineering**, CGPA: 9.13/10* — *Aug 2020 – May 2024*

Work Experience

Applied Researcher — Sep 2025 – May 2026
Do Quantum — *College Park, Maryland, USA*

- Built a modular **Python/Flask post-quantum security prototype** simulating data exchange across hospital, gateway, and pharmaceutical components using **ML-KEM-768, HKDF-SHA-256, AES-256-GCM, and ML-DSA-65**.
- Simulated and tested **ciphertext tampering, random corruption, truncation, and replay-style modification**, verifying that altered encrypted payloads were rejected through signature verification or authenticated decryption.
- Benchmarked cryptographic operations across synthetic **TXT, CSV, JSON, and PDF files from approximately 1 MB to 10 MB**, observing payload-dependent encryption and signature costs while ML-KEM overhead remained comparatively stable.

Technical Skills

Languages: Python, SQL, C++
Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs, Transformers, Diffusion Models, LoRA, CLIP
Generative AI & Retrieval: Hugging Face Transformers, LangGraph, LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, Tool Routing, FAISS, Vector Search, Semantic Search, Sentence-Transformer Embeddings, Cross-Encoder Reranking
Backend & ML Systems: FastAPI, Flask, REST APIs, Async Python, Redis, Model Serving, Inference Pipelines, Model Routing, Circuit Breakers, Backpressure, Rate Limiting, Health Checks, Structured Logging, Load Testing
Cloud, DevOps & Observability: AWS (ECS Fargate, ALB, ECR, SSM, S3), Docker, Terraform, Infrastructure as Code, GitHub Actions, CI/CD, OIDC, Prometheus, Grafana, Autoscaling, Git
Data & Evaluation: Pandas, NumPy, Statistical Analysis, Cross-Validation, Data Augmentation, Model Evaluation, Retrieval Evaluation, LLM-as-Judge, Error Analysis, Gradio

Projects

Inference Reliability Gateway | GitHub — June 2026 – July 2026

- Built an **LLM inference gateway** with **FastAPI, async Python, Redis caching, and complexity-based model routing**, using a common backend interface to separate gateway logic from individual model implementations.
- Implemented **timeouts, retries, circuit breaking, failover, rate limiting, and bounded-concurrency backpressure** to control dependency failures and overload behavior.
- Instrumented the gateway with **Prometheus metrics and Grafana dashboards** and load-tested it with **Locust** to analyze request rate, latency, errors, cache behavior, and in-flight workload under concurrency.
- Measured repeat-request latency of **624 ms cold vs. 5.5 ms cached**, approximately **20× higher throughput** for async simulated I/O versus a blocking baseline, and an estimated **42% lower inference cost** on a 12-prompt mixed workload using model routing.
- Containerized and deployed the gateway on **AWS ECS Fargate** using Docker and Terraform, with ALB, ECR, SSM Parameter Store, GitHub Actions CI/CD via OIDC, and autoscaling configured from **1 to 4 tasks**.

Agentic RAG Research Assistant | GitHub | Live Demo — May 2026 – June 2026

- Built and deployed an **agentic RAG assistant** over **1,953 chunks from 20 AI papers** using LangGraph, FAISS, sentence-transformer embeddings, Groq-hosted Llama 3.3 70B, and Tavily web search.
- Implemented a **LangGraph planner and conditional tool-routing workflow** that directs questions to semantic paper retrieval, live web search, or both before generating a source-grounded response.
- Built an evaluation harness that measured **80% top-1 retrieval precision and 100% recall@5** on a **6-case test set**; diagnosed a cross-encoder reranking regression from **80% to 60% top-1 precision** and replaced the default reranking path with metadata filtering for known retrieval failures.

Multimodal Emotion-to-Art Generation | GitHub — Aug 2025 – Dec 2025

- Built and integrated an end-to-end multimodal pipeline combining a **DistilBERT emotion classifier**, a **27-to-9 NNLS emotion mapping**, and **Stable Diffusion v1.5** for emotion-conditioned image generation.
- Fine-tuned **Stable Diffusion v1.5 with LoRA** and integrated the classifier, mapping layer, diffusion model, and **Gradio** interface into a single inference workflow.
- Evaluated generated images using **CLIP similarity and CLIP-based emotion alignment**, identifying stronger alignment for visually distinct emotions such as anger and disgust than for more abstract emotions such as sadness.

Publications

- **Towards Post-Quantum Secure Pharmacovigilance with ML-KEM and ML-DSA** *(First Author). arXiv preprint*, arXiv:2606.09412 [cs.CR], 2026. DOI: 10.48550/arXiv.2606.09412.
- **Enhanced Communication Security: Leveraging Quantum Cryptography** *(First Author). ICT4SD 2024*, Lecture Notes in Networks and Systems, vol. 1197. Springer, 2025.